

## Short communication

# Poverty, cultural disadvantage and brain development: a study of pre-school children in Mexico

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**Abstract**

Forty-two children, who had been studied previously at the age of 18–30 months, were studied again at 4 years of age. Twenty-two belonged to low socioeconomic strata and were classified as high-risk children (HRC) the other 20 were classified as low-risk children (LRC), and belonged to middle and middle-high socioeconomic strata. Ten minutes of EEG using reference derivations (with linked earlobes) were recorded from each subject. Twenty EEG segments of 3.2 s each were selected by visual inspection for Fourier analysis. Absolute power (AP) was computed for the total EEG energy (1.5–19 Hz) as well as each reference derivation in 4 frequency bands: delta (1.5–3.5 Hz), theta (4–7.5 Hz), alpha (7.5–12.5 Hz) and beta (12.5–19 Hz). HRC had significantly more delta AP than LRC in frontal and central leads, and higher values of theta AP in frontal leads. Alpha AP was higher in LRC in occipital areas and in F8 and T4. This study suggests a maturational lag in HRC. © 1997 Elsevier Science Ireland Ltd.

**Keywords:** EEG frequency analysis; Brain development; Cultural disadvantage; Developmental lag; Poverty; Pre-school children

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**1. Introduction**

Several authors have cited a relationship between poverty, suboptimal cognitive development and academic failure (White, 1982; Patterson et al., 1990; Ramey and Campbell, 1991). However, the mechanisms and processes involved remain poorly understood.

It has also been established that in societies that fail to meet basic human needs (food, health, housing, etc.), widespread biological factors such as malnutrition and infectious diseases often limit children's growth and development (Pollit, 1994).

In addition to the well known negative psychological effects for children produced by an environment characterized by a low level of stimulation, (Bowlby, 1952; Ramey et al., 1975; Bradley and Caldwell, 1976a; Bradley and Caldwell, 1976b; Rutter, 1980; Bendersky and Lewis, 1994) it has been suggested that such an environment also affects EEG maturation (Harmony et al., 1988; Harmony et al., 1990; Otero, 1994). According to this author's working hypothesis, the disadvantage which comes with poverty is

one of the most negative factor producing developmental lags in children.

A previous paper presented a study of broad band EEG frequency analysis of children between 18 and 30 months old (Otero, 1994). In high-risk children, in frontal regions more energy was found in the delta band, and less in alpha, than was the case for low-risk children.

At present the same children are being re-evaluated. This paper presents the results of the HRC and LRC at 4 years of age.

**2. Methods and materials****2.1. Subjects**

Forty-two children were studied, at 4 years of age; 20 were boys and 22 girls. All the children had normal results from neurological and pediatric studies. Twenty-two of the children belonged to low socioeconomic strata and were classified as high-risk children (HRC) through application of a sociocultural questionnaire. The other 20 subjects who were classified as low-risk children (LRC), belonged to

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middle and middle-high socioeconomic strata from the city of Toluca, State of Mexico. They were selected from various nurseries after application of the sociocultural questionnaire, a complete pediatric and neurological examination which included significant personal and family antecedents, and a complete physical examination. This control group was matched by sex and age with the HRC group.

The sociocultural questionnaire was originally designed to classify families from low and very low socioeconomic status (SES). It was originally applied to 500 housewives. The most useful items were obtained by factor analysis. Scores greater than 350 were considered as very low SES, while those scores lower than 300 were considered low SES.

These cut-off points represented one standard deviation above and below the mean value of the 500 subjects. Because all the families of this sample belonged to low social class the distribution was narrow, that's why the separation of both groups (low and very low SES) was made by 50 point separation.

In the present study, the questionnaire was originally applied to 90 families from low, middle and middle-high social classes. LRC mean value for total score was 810.56 (SD, 73.13) and HRC mean value was 292.90 (SD, 12.68). Student's *t*-test was applied for comparing mean differences; *t*-value was 20.47 ( $P < 0.0000$ ). Table 1 illustrate the different items and the maximum score given to each item of the sociocultural questionnaire. Although toxic additions of the parents or other relatives were included as an item in this questionnaire, it was always very low in all families studied.

The HRC had been studied previously (Otero, 1994), at the age of 18–30 months. All the children (HRC and LRC)

were born spontaneously through eutotic labor in hospital conditions, and none showed biological risk antecedents. HRC were followed up first, every 3 months, and later every 6 months, until the moment of testing. All children showed height and weight at the 50th percentile of Mexican Normative Tables (Ramos and Luna, 1964). Neurological examinations revealed no abnormalities or developmental delay.

## 2.2. EEG Recording

EEG recordings were performed in the laboratory during the morning. Ten minutes of EEG using reference derivations, were recorded from F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz and Pz, using linked earlobes (Rodin, 1990). The children were awake, with their eyes closed, in a comfortable reclining chair. A 24 channel digital electroencephalograph MEDICID-3M was used for the recording and further analysis. The amplifier bandwidth was set between 0.5 and 30 Hz. Twenty artefact-free EEG segments, of 3.2 s each, were selected by visual inspection for Fourier analysis. Awake condition of the children was monitored continuously during the study.

## 2.3. EEG analysis

Analysis was done off-line. FFT was performed to calculate the absolute power (AP) for the total EEG energy (1.5–19 Hz) and in 4 frequency bands: delta (1.5–3.5 Hz), theta (4–7.5 Hz), alpha (7.5–12.5 Hz) and beta (12.5–19 Hz). To achieve Gaussian distributions for statistical analysis, the transformations of  $\ln(x)$  for AP was used (John et al., 1980; Gasser et al., 1982; Pollock et al., 1990).

## 2.4. Statistical analysis

With the transformed values of AP in each band and each lead, ANOVAs assessing the main effects of risk and sex, as well as their interactions, were performed. To avoid the problems with 'alpha inflation' using multiple testing there were selected as significant values  $P < 0.005$ .

## 3. Results

Significant differences were observed only with sociocultural risk. There was no effect of sex and no interaction between sociocultural risk and sex.

In frontal and left central leads, the HRC group had significantly more delta AP than the LRC group. The mean values of theta AP were also higher in HRC than in LRC in frontal leads. Although in F3 the *P*-value is 0.008 it was considered because this *P*-value is close to the significant value adopted and because the consistency of the results in frontal areas.

With regard to alpha AP, the highest values were found in

Table 1

Items of psychosocial questionnaire

| Item                                                                        | Score |
|-----------------------------------------------------------------------------|-------|
| Civil status of the parents<br>(married, free union, single mother, etc.)   | 50    |
| Educational level of the parents                                            | 80    |
| Occupation of the parents                                                   | 50    |
| Characteristics of the living environment                                   | 90    |
| Number of relatives and other persons cohabiting<br>with the child          | 50    |
| Family relation<br>(friendly or hostiles relationships between relatives)   | 70    |
| Toxic additions: alcohol, drugs, tobacco, etc. (parents<br>or any relative) | 90    |
| Heavy work during pregnancy                                                 | 90    |
| Behavioral and emotional attitude of the mother to her<br>pregnancy         | 120   |
| Income of the family                                                        | 120   |
| Nutritional habits                                                          | 120   |
| Total 930                                                                   |       |

Children having a score greater than 350 were considered HRC, while those children having a score lower than 300 were considered LRC: mean, 810.56; SD, 73.13. HRC: mean, 300.90; SD, 12.68. LRC vs. HRC: *t*-value = 20.4, *P*-value = 0.0000.

the low risk group, in both occipital regions and in F8 and T4 in alpha band. Total absolute power was significantly higher in frontal, central and parietal regions in HRC. (Fig. 1). Fig. 2 presents a graphic summary of these results.

#### 4. Discussion

The great majority of children from developing countries live in the marginal urban areas of large cities or in rural and semi rural regions. All of these areas are characterized by poverty. There are inadequate sanitary conditions and poor living environments, which have undesirable consequences for children, among them the prevalence of infectious diseases, pregnancy without medical attention and deficiencies in the early care-giving environment. It has been shown that

the early care-giving environment has a powerful influence on the child's subsequent development (Bradley and Caldwell, 1976b; Cravioto and DeLicardie, 1972). Developmental consequences of poverty in regard to cognitive and behavioral aspects have also been the subject of recent studies (Duncan et al., 1994; Campbell and Craig, 1994).

With regard to techniques for measuring relevant brain activity factors, frequency analysis of the EEG has proven to be a precise method for evaluating EEG maturation and EEG background activity (Matousek and Petersén, 1973; John et al., 1980; Watanabe et al., 1984; Harmony, 1989).

In the present paper an analysis of broadband spectral features was performed on EEG referential recordings in order to assess the effect of poverty and culture upon CNS development.

Clear differences were found in AP values. HRC showed

## ABSOLUTE POWER

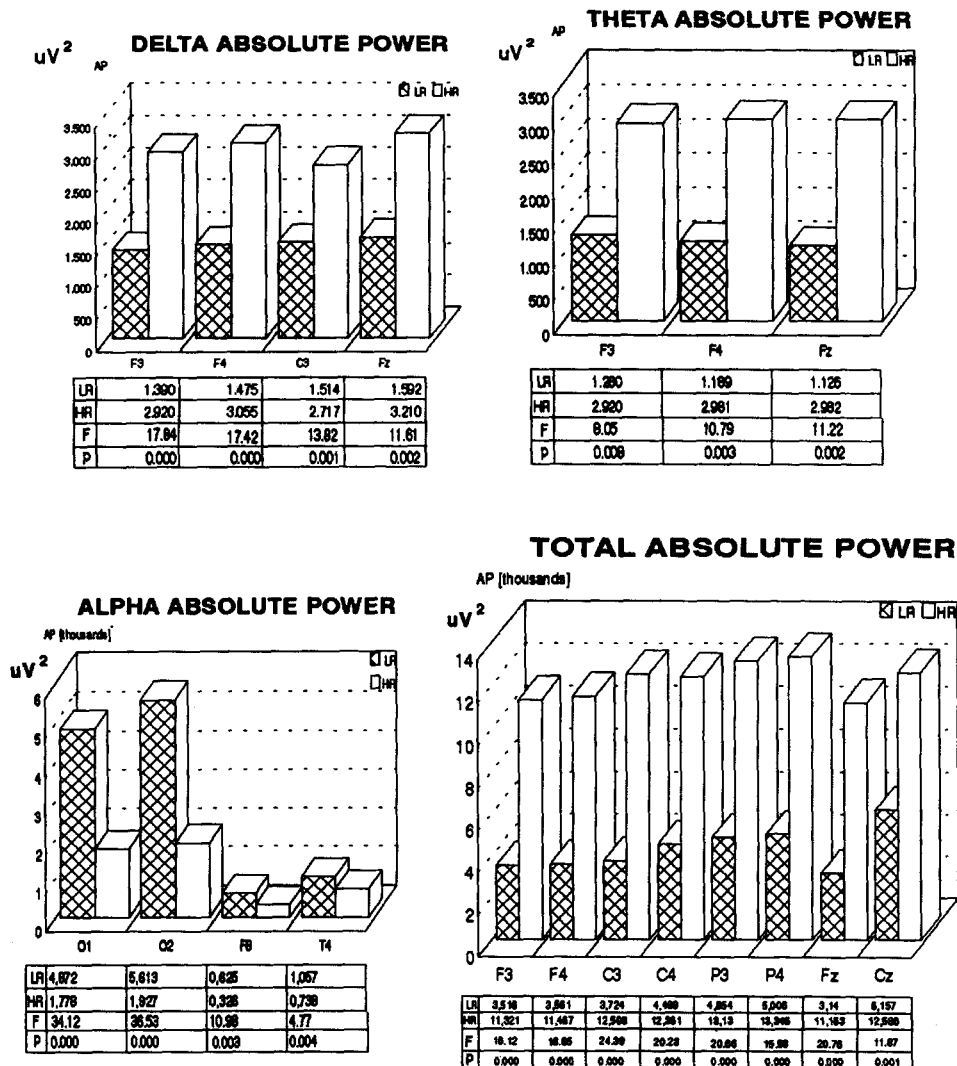


Fig. 1. Absolute power values of delta, theta, alpha bands and of total EEG energy. Tables at the bottom of each graphic show mean, *F*- and *P*-values.

## ABSOLUTE POWER

### Summary

#### HR vs LR

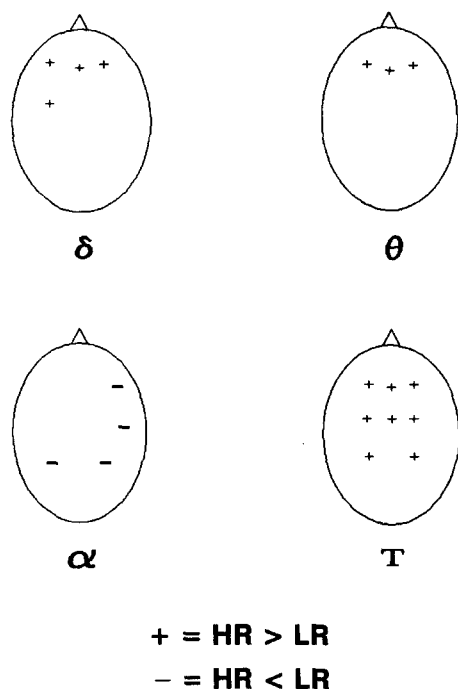


Fig. 2. The plus sign means that, at those places (leads), AP-values were higher for HRC than LRC. Conversely, the minus sign means lower values for HRC than LRC.

more delta and theta power in frontal regions than LRC. On the other hand, HRC showed less power in alpha bands in both occipital areas, F8 and T4. Total AP values showed the same distribution followed by the slow wave bands, i.e., higher in anterior regions where delta and theta bands predominated.

An antecedent of this work was reported previously by Otero (1994), with the same sample of children at 18–30 months of age. Although the results obtained in the 2 studies cannot be directly compared since the recording system was changed, it is possible to perform a qualitative analysis. Taking into account the results of both studies, it is possible to construct the following picture:

- Between 18 and 30 months the main differences in AP values between HRC and LRC were found in frontal leads. Delta power was higher and alpha power lower in the former group. Sex differences were also found.
- When the subject had reached the age of 4, it had become possible for us to use more amplifiers, and a greater number of regions were therefore recorded.

Despite the change in recording system, results were consistent in that delta activity was higher again in HRC in anterior areas, while alpha power were lower in O1, O2, F8, T4.

These results point out to a slower EEG maturation rate in HRC than in LRC. Harmony et al. (1990), studied the development of EEG power-spectrum parameters on children (aged 6–12 years) with different psychosocial characteristics (marginal urban and rural areas vs middle class) from various Latin American countries (Mexico, Venezuela and Cuba) with and without risk antecedents. They concluded that EEG matures at the same rate in children who grew up with adequate nutritional, sanitary and environmental conditions. Psychosocial handicaps were as important as biological risk factors in rural population of children. On the other hand, children with low socioeconomic status (SES) had higher values of AP and higher percentage of delta and lower of alpha than children with good socioeconomic status, the authors suggested that a maturational lag was present in these children.

Early studies found SES to be strongly related to developmental outcomes (Escalona, 1982; Drillien et al., 1980). Van Alstyne (1929) recognized SES to be a distal variable that affects indirectly the child through its impact on such factors as family's lifestyle, availability of resources, etc. Examples of other distal variables are neighborhood and type of residence. Maternal responsivity, availability of appropriate play materials, and opportunities for varied, daily stimulation are considered as proximal variables (they affect directly the child) and have been shown to be related to cognitive abilities (Lewis and Coates, 1980). All of these variables are included in the sociocultural questionnaire used in the present study.

A more unlikely explanation could be a minimal brain organic damage. Although the clinical examination of HRC did not reveal any neurological abnormalities, it could be suggested a light organic brain disorder in HRC. In this sense Fernández et al. (1991) carried out a computed tomography examination in a group of 43 children with normal neurological examination, who showed electrophysiological abnormalities. Six children (with learning disabilities) out of 15 had abnormalities in the computed tomography.

Harmony et al. (1995) carried out a study (broad band frequency analysis of the EEG) with children with severe (reading and writing performance below level for their age and school grade) and minor (performance below level, with minor difficulties) difficulties in reading and writing abilities (RWD). They found more diffuse theta and less alpha powers in RWD children than in control group (adequate performance for age and school grade). Authors interpreted these results as a sign of maturational lag in these children with respect to those with normal performance.

Results of the present study suggest that disadvantages in sociocultural aspects of the families and neighborhoods probably could be associated with a slower EEG maturation

in these children. Of course, despite that clinical examination revealed that HRC did not show significant neurological and/or malnutrition signs, it is not possible to reject the hypotheses that the effects stem from other factors as systemic changes secondary to frequent minor infectious diseases, pregnancy without medical attention, etc.

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